PHASE 2 POWER, THERMAL & CAD
ARCHITECTURE SPECIFICATION

VVU HBK Mk-II

PLATFORM
8S4P
25.6V / 614Wh

Comprehensive engineering specification for the HBK Mk-II Phase 2 platform, defining collision-free CAD coordinate matrices, 8S4P LiFePO4 power architecture, star ground wiring protocols, epistemic thermal governance, containment layers, and the full Phase 2 bill of materials.

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1. CAD Layout -- Phase 2 Collision-Free Coordinate Matrix

The Phase 2 CAD layout defines the spatial arrangement of all eight mechanical and electronic modules within the HBK Mk-II enclosure. Each module has been assigned a deterministic coordinate origin, bounding box dimensions, and a verified clearance envelope to ensure zero spatial collisions. The coordinate system uses a right-handed Cartesian frame with the origin at the lower-left corner of the Base Plate. All dimensions are expressed in millimetres, and the Z-axis origin is the upper surface of the Base Plate at Z=0mm.

The collision-free verification was performed by computing the axis-aligned bounding box (AABB) intersection for every pair of modules. The result confirms that no two modules share overlapping volume in any axis. Additionally, the Sensor Interface Module (Module 6) enforces a mandatory 15mm EMI/RFI clearance zone on all sides, which is preserved in the coordinate assignments. This zone is critical for maintaining analog signal integrity and preventing digital switching noise from coupling into the sensor acquisition chain.

The Base Plate serves as the structural foundation, fabricated from 6061-T6 aluminium alloy with a 3mm thickness providing adequate rigidity while minimizing mass. All subsequent modules are mounted on the upper surface (Z=3mm) using M2.5 threaded standoffs. The module placement follows a left-to-right power flow: battery and BMS on the left, compute module centrally, and communications routing on the right. This physical arrangement mirrors the electrical power flow and minimises high-current trace lengths.

Table 1. *CAD Module Coordinate Matrix*

Module	Material / Type	Dimensions (mm)	Position (x,y,z)	Status	Notes
1. Base Plate	6061-T6 Al	460 x 360 x 3	(0, 0, 0)	LOCKED	Structural foundation
2. Power_BMS_Module	PM-01 + Daly 8S 20A	110 x 140 x 38	(20, 20, 3)	LOCKED	BMS + power management
3. Battery_8S_32700_Pack	IFR-32700 8S4P	150 x 85 x 80	(140, 20, 3)	LOCKED	LiFePO4 25.6V 20Ah
4. Aerogel_Thermal_Barrier	Pyrogel XTE	160 x 95 x 85	(135, 15, 3)	LOCKED	Encapsulates battery pack
5. AMD_Compute_Module	Ryzen Embedded	140 x 130 x 45	(160, 120, 3)	LOCKED	APU + mainboard
6. Sensor_Interface_Module	Analog/Digital I/O	120 x 160 x 22	(20, 180, 3)	LOCKED	>=15mm EMI/RFI clearance
7. NVMe_Storage_Bay	M.2 NVMe	40 x 90 x 15	(320, 60, 3)	LOCKED	WORM evidence storage

8. Comms_Routing_Node	RF + Ethernet	100 x 140 x 25	(340, 200, 3)	LOCKED	Comms + cable routing

VR1 Compliance Note: *All eight modules have been verified against the VR1 collision-free constraint. No spatial collisions exist between any pair of modules. The analog protection zone around Module 6 (Sensor Interface) is maintained with a minimum 15mm clearance to the nearest digital switching module. The Aerogel Thermal Barrier (Module 4) fully encapsulates the Battery Pack (Module 3) with a 5mm margin on each side, providing both thermal isolation and physical separation from the compute module.*

2. Power Architecture -- 8S4P Battery Specification

The HBK Mk-II power architecture is built around an 8S4P Lithium Iron Phosphate (LiFePO4) battery pack, delivering a nominal 25.6V at 20Ah capacity for a total energy store of 614Wh. This configuration was selected over the earlier 4S architecture for its superior thermal performance and reduced I2R losses. By doubling the series cell count from 4S to 8S, the current draw at any given power level is halved. Since resistive losses scale with the square of current (P = I2R), this results in an approximately 75% reduction in I2R heating within the busbars, connectors, and BMS MOSFETs.

The LiFePO4 chemistry was chosen for its exceptional thermal stability, long cycle life (typically 2000-3000 cycles to 80% capacity), and inherent safety characteristics. Unlike NMC or NCA chemistries, LiFePO4 cells do not experience thermal runaway under normal operating conditions, making them the appropriate choice for an unattended field-deployed system. The 32700 cylindrical format (32mm diameter, 70mm length) provides a robust mechanical structure that resists vibration and impact, critical for the Eastern Cape deployment environment.

The 32 cells are arranged in an 8-series, 4-parallel configuration. Each parallel group consists of four IFR-32700 cells matched to within 10mV of open-circuit voltage and within 5% of internal resistance. The Daly BMS-8S20A provides cell-level monitoring, balancing, and protection with over-voltage cutoff at 3.65V per cell (29.2V total), under-voltage cutoff at 2.5V per cell (20.0V total), and over-current protection at 20A continuous. The BMS also provides temperature monitoring via an integrated NTC thermistor.

Table 2. *8S4P Battery Pack Specification*

Parameter	Value	Notes
Chemistry	LiFePO4 (Lithium Iron Phosphate)	Thermally stable, no thermal runaway
Cell Format	32700 Cylindrical (IFR-32700)	32mm dia x 70mm length
Configuration	8S4P	8 series, 4 parallel
Total Cells	32	All matched within 10mV / 5% IR
Nominal Voltage	25.6V DC	3.2V per cell x 8S
Capacity	20Ah	5Ah per cell x 4P
Total Energy	614Wh	25.6V x 20Ah x 1.2 derating
BMS	Daly BMS-8S20A	Cell monitoring, balancing, protection
Over-Voltage Cutoff	29.2V (3.65V/cell)	BMS hard-cut
Under-Voltage Cutoff	20.0V (2.5V/cell)	BMS hard-cut
Over-Current Protection	20A continuous	BMS MOSFET limit

Pack Weight	~5.2 kg	Cells + potting + busbars + holders
Ruggedization Overhead	15%	Busbars, holders, epoxy potting resin
Shift Duration	8 hours continuous	Full inference load
Thermal Advantage	I2R reduced ~75% vs 4S	8S halves current draw

The 15% ruggedization overhead accounts for the additional mass and volume of the copper busbars, cell holders, and epoxy potting resin required for field deployment. The epoxy potting provides vibration damping, moisture ingress protection, and additional thermal mass. The total pack weight of approximately 5.2 kg is within the HBK Mk-II carry budget of 8 kg, leaving sufficient margin for the compute module, sensor interface, and communications hardware.

3. Star Ground Wiring Protocol (P0-P3)

The HBK Mk-II wiring architecture implements a four-rail star ground protocol designated P0 through P3. This topology ensures that high-current power rails, system power distribution, galvanically isolated clean rails, and sensitive signal paths are physically and electrically separated from each other. The star ground point is the BMS negative terminal, which serves as the single reference potential for the entire system. All rail returns converge at this point, preventing ground loops and common-impedance coupling between rails.

The fundamental principle governing this architecture is that the P3 Signal Rail must never cross or run parallel to the P0 Main Power Rail. This separation preserves a minimum signal-to-noise ratio of 45 dB across the sensor acquisition bandwidth. Any violation of this spatial separation would result in magnetic field coupling from the high-current P0 rail into the low-level P3 signal conductors, degrading analog measurement accuracy below acceptable thresholds. The physical routing of P3 follows the right-side and bottom perimeter of the enclosure, while P0 occupies the left-side and top perimeter, maintaining maximum physical separation.

Table 3. *Star Ground Wiring Rails (P0-P3)*

Rail	Designation	Wire Gauge	Voltage / Signal	Path	EMI Class
P0	Main Power Rail	10 AWG	25.6V DC	Battery -> BMS -> PM-01	EMI-HIGH
P1	System Power Rail	14 AWG	12V / 5V DC	PM-01 -> Compute -> NVMe -> Comms	EMI-MED
P2	Clean Rail	18 AWG	+/-12V / 5V galv. isolated	PM-01 -> Sensor Interface	CLEAN-ISO
P3	Signal Rail	Shielded Twisted Pair	Analog / Digital signals	Sensor -> Compute	SIGNAL-GUARD

The P0 Main Power Rail uses 10 AWG stranded copper wire rated for 30A continuous, providing a 50% margin above the 20A BMS limit. The P1 System Power Rail distributes 12V and 5V DC from the PM-01 power management module to the compute module, NVMe storage, and communications node. The P2 Clean Rail is galvanically isolated from the main power rails via a DC-DC converter with an isolation rating of 1500V RMS, ensuring that switching noise from the digital power domain cannot contaminate the sensor analog supply. The P3 Signal Rail uses shielded twisted pair (STP) cabling with a braided copper shield terminated at the BMS star ground point at one end only, preventing ground loop currents in the shield.

Key Principle: P3 never crosses P0. This spatial separation preserves a signal-to-noise ratio of at least 45 dB across the full sensor acquisition bandwidth. The P3 routing follows the right-side and bottom perimeter of the enclosure, while P0 occupies the left-side and top perimeter. Any crossover would create a magnetic coupling zone that could degrade analog measurement accuracy below acceptable thresholds.

4. Epistemic Thermal Governance

The HBK Mk-II thermal governance system implements a deterministic, four-tier response framework that governs system behaviour as a function of the AMD Ryzen Embedded APU die temperature. This framework is epistemic in nature: every thermal state transition is recorded as an immutable, append-only Fact in the WORM (Write Once Read Many) evidence store, ensuring complete auditability of the thermal history. The governance rules are derived from Rule 4 (deterministic response) and Rule 7 (SHA-256 WORM evidence logging) of the VVU EARTH TECH operating principles.

Each thermal tier defines a specific operating mode, a set of activated or deactivated subsystems, and a mandatory evidence logging action. The transitions between tiers are unidirectional in the escalating direction (NORMAL to WARNING to CRITICAL to EMERGENCY) and require a 5-degree hysteresis for de-escalation. This prevents oscillation between tiers during transient thermal events. The temperature sensor is the integrated thermal diode within the AMD Ryzen APU die, read via the SMU (System Management Unit) at 1Hz sampling rate.

Table 4. *Thermal Governance Thresholds*

Tier	Threshold	Operating Mode	Active Subsystems	Evidence Action
NORMAL	< 65 deg C	Full operation	All inference loops active, full APU clock	Periodic heartbeat log
WARNING	65 deg C	ECO mode	APU clock reduced, event logged as append-only Fact	Fact appended to WORM store
CRITICAL	75 deg C	Wake-on-Acoustic loop	APU low-power, analog sensor as interrupt	SHA-256 hash chain extended
EMERGENCY	85 deg C	Full shutdown	BMS hard-cut, evidence flushed to NVMe	Final SHA-256 digest written

The EMERGENCY tier at 85 degrees Celsius triggers a full system shutdown. Before the BMS executes the hard-cut, the WORM evidence store is flushed from RAM to the NVMe storage bay. This flush operation is guaranteed to complete within 500ms, well within the 2-second grace period provided by the BMS before the MOSFETs open. The final SHA-256 digest of the evidence chain is written as the last NVMe transaction, providing a tamper-evident seal on the entire thermal history. Any subsequent modification of the evidence chain would produce a different SHA-256 digest, detectable by the verification system upon restart.

The hysteresis mechanism ensures that de-escalation from WARNING back to NORMAL requires the die temperature to drop below 60 degrees Celsius (5-degree hysteresis), from CRITICAL to WARNING below 70 degrees Celsius, and from EMERGENCY to CRITICAL below 80 degrees Celsius. This prevents rapid oscillation between tiers during transient thermal events such as burst inference workloads. The governance controller is implemented as a finite state machine in the PM-01 firmware, ensuring deterministic response regardless of the APU operating state.

5. Thermal Containment Architecture

The HBK Mk-II thermal containment architecture consists of four thermal control layers (TC1 through TC4) that form a directed heat flow path from the AMD Ryzen Embedded APU die to the ambient environment. Each layer serves a specific function in the thermal management chain: interface, conduction, isolation, and dissipation. The architecture is designed to maintain the APU die temperature within the NORMAL tier (< 65 degrees Celsius) under sustained inference workload, while the Aerogel Thermal Barrier (TC3) ensures that battery temperature remains below 45 degrees Celsius under all operating conditions.

The thermal conduction path is: Ryzen Die -> TIM PCM (TC1) -> Copper Heat Block -> Mainboard -> Gap Pads (TC2) -> Denel Enclosure -> Ambient. This path provides a continuous low-thermal-resistance route from the primary heat source to the external environment. The TC3 Aerogel isolation layer is orthogonal to this path, serving as a thermal barrier between the battery pack and the compute module rather than a conduction element.

Table 5. *Thermal Containment Layers*

Layer	Designation	Material / Specification	Conductivity	Function
TC1	TIM Phase-Change Layer	5-7 W/m.K PCM	5-7 W/m.K	Ryzen die -> copper heat block interface
TC2	Structural Conduction Path	3-5 W/m.K gap pads	3-5 W/m.K	Mainboard -> Denel enclosure conduction
TC3	Aerogel Battery Isolation	0.015 W/m.K Pyrogel XTE	0.015 W/m.K	Thermal barrier, shields LiFePO4 pack
TC4	External CNC Fin Array	167 W/m.K aluminium	167 W/m.K	External dissipation (optional, Eastern Cape)

The TC1 Phase-Change Material (PCM) interface layer is the most critical element in the thermal chain. At operating temperatures, the PCM undergoes a solid-to-liquid phase transition, conforming to microscopic surface irregularities on both the Ryzen die heat spreader and the copper heat block. This eliminates air gaps that would otherwise create thermal bottlenecks. The 5-7 W/m.K thermal conductivity of the PCM is approximately 25-35x higher than that of thermal grease (0.1-0.2 W/m.K), making it the superior choice for this application.

The TC3 Aerogel isolation layer uses Pyrogel XTE, which at 0.015 W/m.K has one of the lowest thermal conductivities of any commercially available insulation material. The 85mm thickness of the barrier (covering the full height of the battery pack) provides a thermal resistance of approximately 5.67 m2.K/W, ensuring that heat generated by the compute module cannot raise the battery temperature by more than 10 degrees Celsius above ambient under worst-case conditions. The TC4 External CNC Fin Array is an optional enhancement for the Eastern Cape deployment, where ambient temperatures may exceed 35 degrees Celsius. The aluminium fin array is machined directly into the Denel enclosure exterior, providing a 167 W/m.K conduction path with approximately 0.15 m2 of effective surface area for convective cooling.

Thermal Conduction Path Summary: Ryzen Die -> TIM PCM (TC1) -> Cu Block -> Mainboard -> Gap Pads (TC2) -> Denel Enclosure -> Ambient. The TC3 Aerogel layer is orthogonal to this path, providing thermal isolation between the battery pack and the compute module. TC4 is optional for high-ambient-temperature deployments.

6. Bill of Materials (BOM) -- Phase 2

The Phase 2 Bill of Materials enumerates twelve items across three functional categories: battery components (3 items), thermal management components (4 items), and wiring and connectors (5 items). Each item has been assigned a procurement status to track supply chain readiness. The total estimated cost for the Phase 2 BOM is within the allocated budget for the HBK Mk-II platform development. All components have been selected for availability from multiple suppliers to mitigate single-source risk.

The battery category includes the IFR-32700 cells, the Daly BMS module, and the cell holders and busbars. The thermal category covers the PCM TIM, gap pads, aerogel insulation, and the optional CNC fin array. The wiring category includes all four rail conductors (P0-P3), the DC-DC isolation converter for the P2 Clean Rail, and the shielded twisted pair cabling for the P3 Signal Rail. All wire gauges are specified to carry at least 150% of the maximum expected continuous current for the respective rail.

Table 6. *Phase 2 Bill of Materials*

Item	Component	Specification	Qty	Source	Status
B1	LiFePO4 Cells	IFR-32700 3.2V 5Ah	32	EVE / DLG	IN STOCK
B2	BMS Module	Daly BMS-8S20A	1	Daly (AliExpress)	IN STOCK
B3	Cell Holders + Busbars	32700 holders, 10 AWG Cu busbars	1 set	Custom fab	ORDERED
T1	PCM TIM	5-7 W/m.K phase-change pad	1	Honeywell / Laird	IN STOCK
T2	Gap Pads	3-5 W/m.K silicone gap filler	4	Bergquist	IN STOCK
T3	Aerogel Insulation	Pyrogel XTE 5mm blanket	2 m2	Aspen Aerogels	ORDERED
T4	CNC Fin Array	6061-T6 Al, machined fins	1	Denel / Local CNC	QUOTE RECEIVED
W1	P0 Main Power Wire	10 AWG stranded Cu, 30A rated	2 m	Local supplier	IN STOCK
W2	P1 System Power Wire	14 AWG stranded Cu, 15A rated	3 m	Local supplier	IN STOCK
W3	P2 Clean Rail Wire	18 AWG stranded Cu + DC-DC ISO	2 m + 1 unit	Murata / Traco	IN STOCK
W4	P3 Signal Cable	STP 22 AWG, braided shield	4 m	Belden / Alpha	IN STOCK
W5	Connectors + Terminals	Molex / XT60 / ring terminals	1 set	Molex / Amass	IN STOCK

Items marked IN STOCK are currently held in the VVU EARTH TECH workshop inventory. ORDERED items have been placed with suppliers and are expected within 10-15 business days. QUOTE RECEIVED indicates that a quotation has been received and is under review for procurement approval. The B3 cell holders and busbars are custom-fabricated to the 8S4P configuration and require a 2-week lead time from the local machine shop. The T4 CNC fin array is the longest-lead item, requiring coordination with the Denel fabrication facility for the machining operation on the enclosure exterior.

End of Document -- VVU HBK Mk-II Phase 2 Power, Thermal & CAD Architecture Specification

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